

TelcoBot — LangGraph Workflow Diagram

Architecture Overview

```
graph TD
    A[START<br/>User Message] --> B[Intent Classifier<br/>LLM classifies intent]

    B -->|billing| C[Billing Agent<br/>Bills, payments, balance]
    B -->|plans| D[Plans Agent<br/>Plan info, comparisons]
    B -->|technical| E[Technical Agent<br/>Troubleshooting, RAG search]
    B -->|usage| F[Usage Agent<br/>Data, calls, SMS tracking]
    B -->|account| G[Account Agent<br/>Profile, tickets, lookup]
    B -->|general| H[General Agent<br/>Greetings, help menu]

    C -->|tool_calls| I[Tool Executor<br/>Runs tool functions]
    D -->|tool_calls| I
    E -->|tool_calls| I
    F -->|tool_calls| I
    G -->|tool_calls| I

    I -->|tool_results| C
    I -->|tool_results| D
    I -->|tool_results| E
    I -->|tool_results| F
    I -->|tool_results| G

    C -->|response| J[END<br/>Return AI response]
    D -->|response| J
    E -->|response| J
    F -->|response| J
    G -->|response| J
    H -->|response| J

    style A fill:#4CAF50,stroke:#333,color:#fff
    style B fill:#2196F3,stroke:#333,color:#fff
    style C fill:#FF9800,stroke:#333,color:#fff
    style D fill:#9C27B0,stroke:#333,color:#fff
    style E fill:#F44336,stroke:#333,color:#fff
    style F fill:#00BCD4,stroke:#333,color:#fff
    style G fill:#795548,stroke:#333,color:#fff
    style H fill:#607D8B,stroke:#333,color:#fff
    style I fill:#FFC107,stroke:#333,color:#000
    style J fill:#f44336,stroke:#333,color:#fff
```

Tool Mapping

```
graph LR
  subgraph "💰 Billing Tools"
    BT1[get_bill_details]
    BT2[get_payment_history]
    BT3[get_outstanding_balance]
  end

  subgraph "📋 Plan Tools"
    PT1[list_available_plans]
    PT2[get_plan_details]
    PT3[compare_plans]
  end

  subgraph "🔧 Tech Tools"
    TT1[search_knowledge_base<br/>RAG + ChromaDB]
  end

  subgraph "📊 Usage Tools"
    UT1[get_usage_summary]
    UT2[get_daily_usage]
  end

  subgraph "👤 Account Tools"
    AT1[get_customer_profile]
    AT2[lookup_customer_by_name]
    AT3[create_support_ticket]
    AT4[get_open_tickets]
  end
```

Data Flow

```
sequenceDiagram
  participant U as 🧑 User
  participant G as 🗣️ Gradio UI
  participant C as 🧠 Classifier
  participant A as 🤖 Agent
  participant T as 🛠️ Tools
  participant DB as 🗄️ SQLite
  participant VS as 🌐 ChromaDB

  U->>G: Types message
  G->>C: Forward with session memory
  C->>C: Classify intent (LLM)
  C->>A: Route to specialised agent
  A->>T: Call tool(s) if needed
  T->>DB: Query customer data
  DB-->>T: Return results
  T->>VS: Search knowledge base (RAG)
  VS-->>T: Return relevant docs
  T-->>A: Tool results
  A->>A: Generate response (LLM)
  A-->>G: AI response
  G-->>U: Display response
```

Memory Architecture

```

graph LR
  subgraph "Session Memory (MemorySaver)"
    M1[Turn 1: User msg + AI response]
    M2[Turn 2: User msg + Tool calls + AI response]
    M3[Turn 3: User msg + AI response]
    M1 --> M2 --> M3
  end

  subgraph "Thread Management"
    T1[Session UUID → Thread ID]
    T2[Clear Chat → New UUID]
  end

  T1 --> M1
  T2 -. ->|reset| M1

```